

E-COMMERCE COMPANY CASE STUDY

Executive Analytics Report & Strategic Business Deliverables | Data Analytics Bootcamp Evaluation

1. Executive Overview & Case Study Context

As a Data Analyst evaluating company performance, this study analyzes multi-table transaction data spanning **March 2023 to February 2024** (200 orders, 100 registered customers, 8 catalog products). The analysis addresses core operational challenges across four key business verticals identified in the case study problem statement:

- Customer Insights:** Mapping geographical revenue distribution across 10 major Indian cities, profiling VIP buyers, segmenting purchasing frequency, and identifying inactive accounts (zero purchase history).
- Product Analysis:** Evaluating catalog revenue drivers vs. volume movement, assessing category-level revenue share, and mining cross-selling product pairs to optimize bundling.
- Sales Optimization:** Tracking monthly and quarterly revenue trajectories, identifying seasonal demand peaks, analyzing Average Order Value (AOV), and segmenting basket sizes.
- Inventory Management:** Calculating monthly consumption rates (run-rates), classifying inventory velocity (Fast vs. Slow-Moving), and establishing 90-day restocking recommendations.

TOTAL REVENUE Rs 19,783,000 Across 200 Orders	TOTAL UNITS SOLD 1,033 Units 8 Catalog Products	AVERAGE ORDER VALUE Rs 98,915 Per Transaction	ACTIVE BUYER RATE 84.0% 84 Active / 16 Inactive
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2. Customer Insights & Regional Performance

Analysis of customer geographic distribution shows strong market concentration in southern and northern metropolitan hubs. **Chennai** leads all locations with **Rs 3,890,000** (19.7% of total revenue), followed by **Lucknow** (Rs 2,206,000) and **Jaipur** (Rs 2,102,000). Out of 100 registered customers, 84 have made purchases while 16 customers remain completely inactive.

Location	Reg. Cust.	Active Cust.	Total Orders	Total Revenue (INR)	AOV (INR)
Chennai	15	15	34	Rs 3,890,000	Rs 114,411
Lucknow	9	9	24	Rs 2,206,000	Rs 91,916
Jaipur	11	11	22	Rs 2,102,000	Rs 95,545
Delhi	16	16	28	Rs 2,066,000	Rs 73,785
Kolkata	8	8	22	Rs 1,928,000	Rs 87,636
Bangalore	8	8	15	Rs 1,899,000	Rs 126,600
Other Cities (4)	33	33	55	Rs 5,692,000	Rs 103,491

Top VIP Customer Segment:

The top 3 spenders on the platform are **Romil Bora** (Chennai, Rs 889,000 across 6 orders), **Rhea Issac** (Jaipur, Rs 864,000 across 4 orders), and **Indrans Lanka** (Lucknow, Rs 713,000 across 5 orders). The top 10 customers contribute Rs 6.15M (~31.1% of company revenue), highlighting the critical importance of a dedicated VIP loyalty program.

3. Product Performance & Category Share

The catalog comprises 8 products across 3 categories. **Laptop 15" Pro** is the primary revenue driver at **Rs 7,560,000** (38.2% share), while **Digital SLR Camera** leads volume among high-margin goods with **151 units sold** (Rs 6,040,000). Category-wise, Electronics dominates with 64.5% revenue share, followed by Photography (30.5%) and Wearable Tech (5.0%).

ID	Product Name	Category	Price (INR)	Units Sold	Total Revenue (INR)	Share (%)
2	Laptop 15" Pro	Electronics	Rs 60,000	126	Rs 7,560,000	38.22%
7	Digital SLR Camera	Photography	Rs 40,000	151	Rs 6,040,000	30.53%

1	Smartphone 6"	Electronics	Rs 15,000	108	Rs 1,620,000	8.19%
4	E-Book Reader	Electronics	Rs 12,000	130	Rs 1,560,000	7.89%
3	Bluetooth Headphones	Electronics	Rs 8,000	135	Rs 1,080,000	5.46%
6	Portable Speaker	Electronics	Rs 7,000	134	Rs 938,000	4.74%
5	Smartwatch Tracker	Wearable Tech	Rs 5,000	119	Rs 595,000	3.01%
8	Wireless Earbuds	Wearable Tech	Rs 3,000	130	Rs 390,000	1.97%

Cross-Selling & Basket Co-Purchase Insights:

Basket analysis reveals frequent co-purchase pairings: **Wireless Earbuds + E-Book Reader** (31 joint orders), **E-Book Reader + Laptop 15" Pro** (29 joint orders), and **Bluetooth Headphones + Wireless Earbuds** (28 joint orders). Product bundling during checkout presents immediate opportunities to lift basket size.

4. Sales Optimization & Seasonal Trajectory

Monthly revenue shows strong seasonality. Peak sales occurred in **September 2023 (Rs 2,927,000)** and **December 2023 (Rs 2,774,000)**, coinciding with national festival shopping and end-of-year holiday promotions. Conversely, sales troughed in **February 2024 (Rs 396,000)**. High-value orders (> Rs 100,000) account for 69.2% of overall platform revenue.

Month	Orders	Monthly Revenue (INR)	AOV (INR)	Monthly Performance Trend
2023-03	13	Rs 789,000	Rs 60,692	Baseline Start
2023-04	21	Rs 1,704,000	Rs 81,143	Spring Surge
2023-05	18	Rs 1,582,000	Rs 87,889	Steady Demand
2023-07	26	Rs 2,568,000	Rs 98,769	Summer Peak
2023-09	24	Rs 2,927,000	Rs 121,958	Highest Monthly Peak (Festive)
2023-12	21	Rs 2,774,000	Rs 132,095	Year-End Holiday Peak
2024-02	9	Rs 396,000	Rs 44,000	Post-Holiday Trough

5. Inventory Management & Demand Forecasting

Based on a 12-month operating window, catalog products were analyzed for unit consumption velocity. **Digital SLR Camera** (12.6 units/mo) and **Bluetooth Headphones** (11.3 units/mo) exhibit the highest inventory turnover. Recommended 90-day restocking quantities include a **20% safety stock buffer** to eliminate stockout risks during demand surges.

Product Name	Category	Annual Sales	Monthly Run-Rate	Velocity Status	Rec. 90-Day Restock (20% Buffer)
Digital SLR Camera	Photography	151 units	12.6 units/mo	Fast Moving (High Risk)	45 units
Bluetooth Headphones	Electronics	135 units	11.3 units/mo	Fast Moving	41 units
Portable Speaker	Electronics	134 units	11.2 units/mo	Fast Moving	40 units
E-Book Reader	Electronics	130 units	10.8 units/mo	Fast Moving	39 units
Wireless Earbuds	Wearable Tech	130 units	10.8 units/mo	Fast Moving	39 units
Laptop 15" Pro	Electronics	126 units	10.5 units/mo	Fast Moving (High Value)	38 units
Smartwatch Tracker	Wearable Tech	119 units	9.9 units/mo	Moderate Moving	36 units
Smartphone 6"	Electronics	108 units	9.0 units/mo	Moderate Moving	32 units

6. Actionable Business Recommendations

1. **Regional Growth Strategy:** Concentrate digital marketing and regional fulfillment centers in Chennai, Lucknow, and Jaipur. Chennai alone accounts for nearly 20% of sales; dedicated regional warehouses will lower transit times and shipping costs.
2. **Inactive Account Activation:** Deploy automated email/SMS onboarding flows with a Rs 500 first-order voucher for the 16 un-activated customer accounts to improve conversion rates.

- 3. Automated Cross-Selling:** Implement dynamic bundle prompts at checkout (e.g., offering Wireless Earbuds at a 15% discount when purchasing a Laptop 15" Pro or E-Book Reader) to capitalize on established co-purchase patterns.
- 4. Demand-Driven Procurement:** Issue quarterly POs aligned with projected run-rates (e.g., 38 Laptops, 45 Cameras per quarter) with suppliers to avoid stockouts ahead of September and December peak seasons.